

# Md. Ajmain Istiak Apon

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## EDUCATION

### University of Dhaka

B.Sc. in Robotics and Mechatronics Engineering

*Relevant Coursework:* Machine Learning, Artificial Intelligence, Intelligent Systems and Robotics, Digital Image Processing and Robot Vision, Digital Signal Processing, Linear Algebra, Statistics for Engineers.

Dhaka, Bangladesh

Expected Oct 2026

## EXPERIENCE

### Undergraduate Research Assistant

2024 – 2025

University of Dhaka — Funded Research Project: Stomata Detection with YOLOv11

- Trained a **YOLOv11n** model to detect and localize stomata in microscopy images of plant leaves, achieving **0.992 mAP@0.5** and **0.836 mAP@0.5:0.95** (precision 0.969, recall 0.966).
- Curated a custom dataset of **300+ images** annotated in **Roboflow**; built the training pipeline with augmentation, train/validation split, and held-out evaluation.

## RESEARCH & PROJECTS

### Senior Thesis — Generative Memorability

Jan 2026 – Present

*Steering Latent Diffusion Models via Graph-Embedded Spatial Guidance — under faculty supervision*

- Integrated AMNet memorability guidance into a Latent Diffusion Model (Stable Diffusion) pipeline to steer generation toward more memorable outputs while preserving fidelity.
- Achieved **+60% memorability uplift** across a 20-prompt evaluation set, independently validated with **ResMem** as an external scorer.
- Extending the pipeline with graph-embedded spatial guidance via custom U-Net modifications and cross-attention steering, benchmarked against Stable Diffusion baselines.

### Zero-Shot Medical World Models (Z-MWM)

🔗 ajmainapon/SSLR

*Independent Research Project — adapting Stanford's Zero-Shot World Model framework from video to volumetric CT segmentation Jan 2026 – Present*

- Pretraining **ViT-Base/16** with an **IJEPA-style** predictor over 2.5D slice triplets ( $S_{z-k}, S_z$ ) on **278K+ slices** from 1,228 unlabeled TotalSegmentator volumes, decoupling tissue appearance from anatomical geometry without labels.
- Engineered the predictor to a 4-layer Transformer with a learnable mask token and EMA target encoder (momentum 0.996) after a shallow-MLP variant collapsed within 100 steps; settled on **75% sparse masking** and smooth-L1 loss in the latent space.
- Stabilized training on a single **24GB RTX 3090 Ti** with **bfloat16** mixed precision, gradient clipping (max-norm 1.0), and gradient accumulation (effective batch 64); achieving 17 min/epoch and a stable 0.06–0.09 IJEPA loss plateau over 40 epochs.
- Targeting **zero-shot organ segmentation via causal response mapping** (perturbing latent patches and reading the ripple across neighbors), with a few-shot frozen-ViT +  $1 \times 1$  Conv decoder Dice baseline (3–5 annotated volumes) as a publication safety net vs. nnU-Net / MedSAM. *Stack:* PyTorch, timm, MONAI, Nibabel, CUDA.

### REST API with JWT Auth (FastAPI)

🔗 ajmainapon/FASTAPI-Project

*Personal Project*

Apr 2026

- Built a multi-router FastAPI service for an articles/authors blog backend with **SQLAlchemy** ORM and **Pydantic v2** request/response schemas.
- Implemented **OAuth2 password-flow** login issuing **JWT bearer tokens** (bcrypt password hashing via passlib); structured the app with dependency-injected DB sessions and per-resource routers.

## TECHNICAL SKILLS

**Languages** Python, C++, C, SQL, JavaScript

**ML / CV** PyTorch, timm, TensorFlow, Scikit-learn, Hugging Face, OpenCV, MONAI

**LLM / API** LangChain, LangGraph, FastAPI, SQLAlchemy, Pydantic, OAuth2 / JWT

**Tools** Git, Linux, CUDA, ROS2, STM32

## COMPETITIVE PROGRAMMING

**Codeforces (fynm):** 500+ problems solved across dynamic programming, graph theory, greedy, and advanced data structures.

## LEADERSHIP

• **President**, Robotics & Mechatronics Student Club, University of Dhaka

Oct 2025 – Present

• **Program Coordinator**, IEEE Robotics & Automation Society, University of Dhaka

Mar 2024 – Apr 2025